

Power stage development for a dual active bridge DC DC power converter

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Abstract: The DAB power stage uses two active H-bridges and a high frequency transformer with leakage inductance as the main energy transfer element between DC ports. We model, tune, and compare single phase shift (SPS) and dual phase shift (DPS) modulation. Open and closed loop tests track waveforms, soft switching, and reactive current. Overall, SPS is simpler, DPS extends the ZVS analysis, cuts reactive stress.

1. INTRODUCTION

❖ Modern power systems need compact, isolated, and bidirectional DC-DC conversion to connect batteries, DC buses, and fast charging ports. The Dual Active Bridge (DAB) is a strong fit because its power stage two active H-bridges and a high frequency transformer moves energy through the transformer's leakage inductance with simple, fixed switching frequency. By commanding the phase between the bridge voltages, the converter sets both the direction and amount of power flow while enabling soft switching. Validating the DAB power stage, starting with single phase shift (SPS) control as a clean baseline and then introducing dual phase shift (DPS) to reshape inductor current, cut reactive components, and widen the zero voltage switching window. Matlab Simulink studies compare the behavior, quantify current stress.

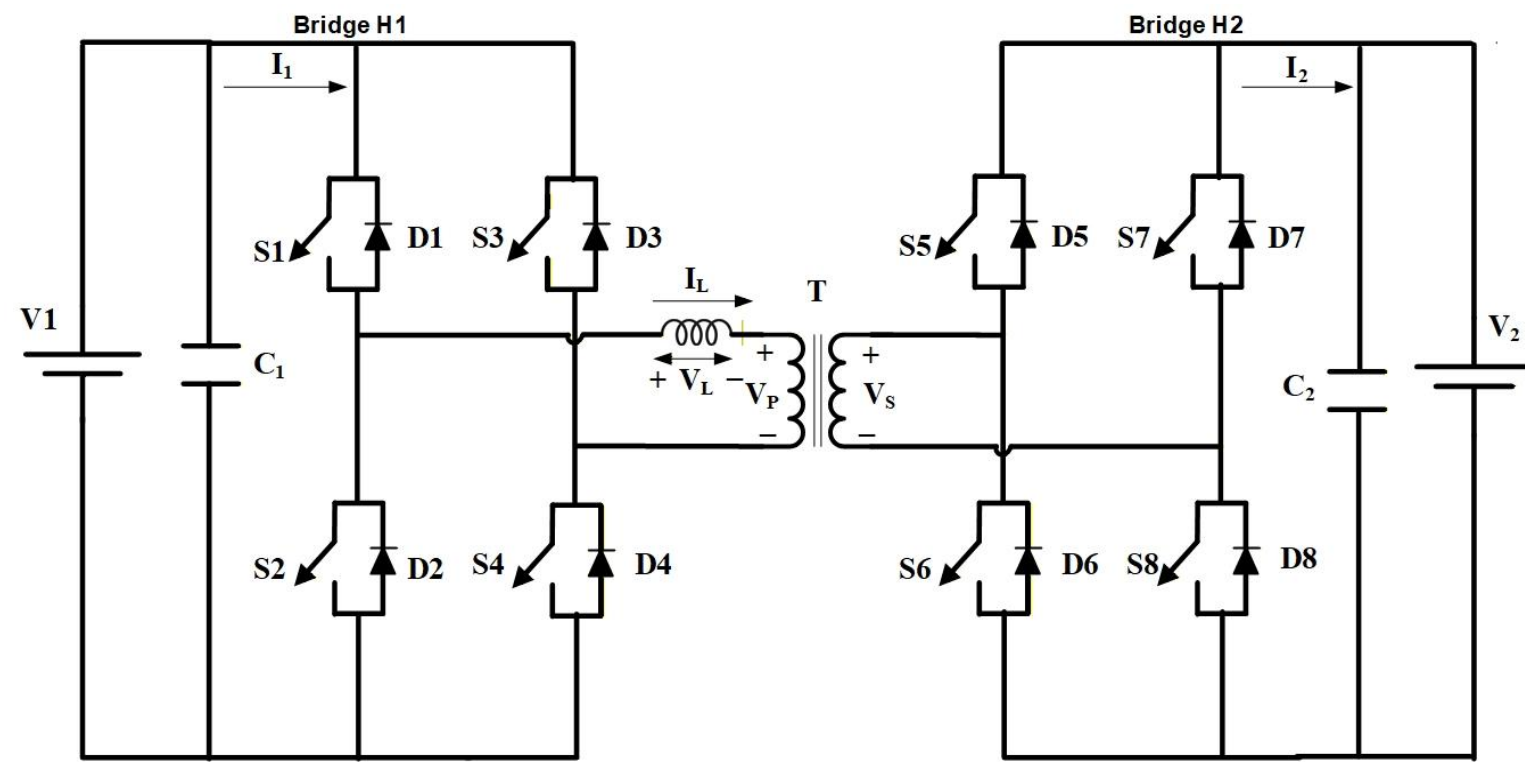


Figure 1: Dual Active Bridge Converter

2. MOTIVATION AND PROBLEM FORMULATION

- ❖ The power stage of a dual active bridge converter transfers energy between two dc ports. In practice, SPS is attractive for simplicity but produces reactive current, raising rms current especially at light load. If dead time is missing loss of ZVS, high switching loss. Our objectives are to validate the plant, compare SPS against DPS, provide dead time so it enables soft switching, minimize reactive power and implement closed loop regulation under dynamic conditions.
- ❖ The first goal was to examine the power stage for a two winding dual active bridge converter as shown in figure 1, where as the second goal is to examine the power stage for a three winding quad port dual active bridge converter as shown in figure 2.

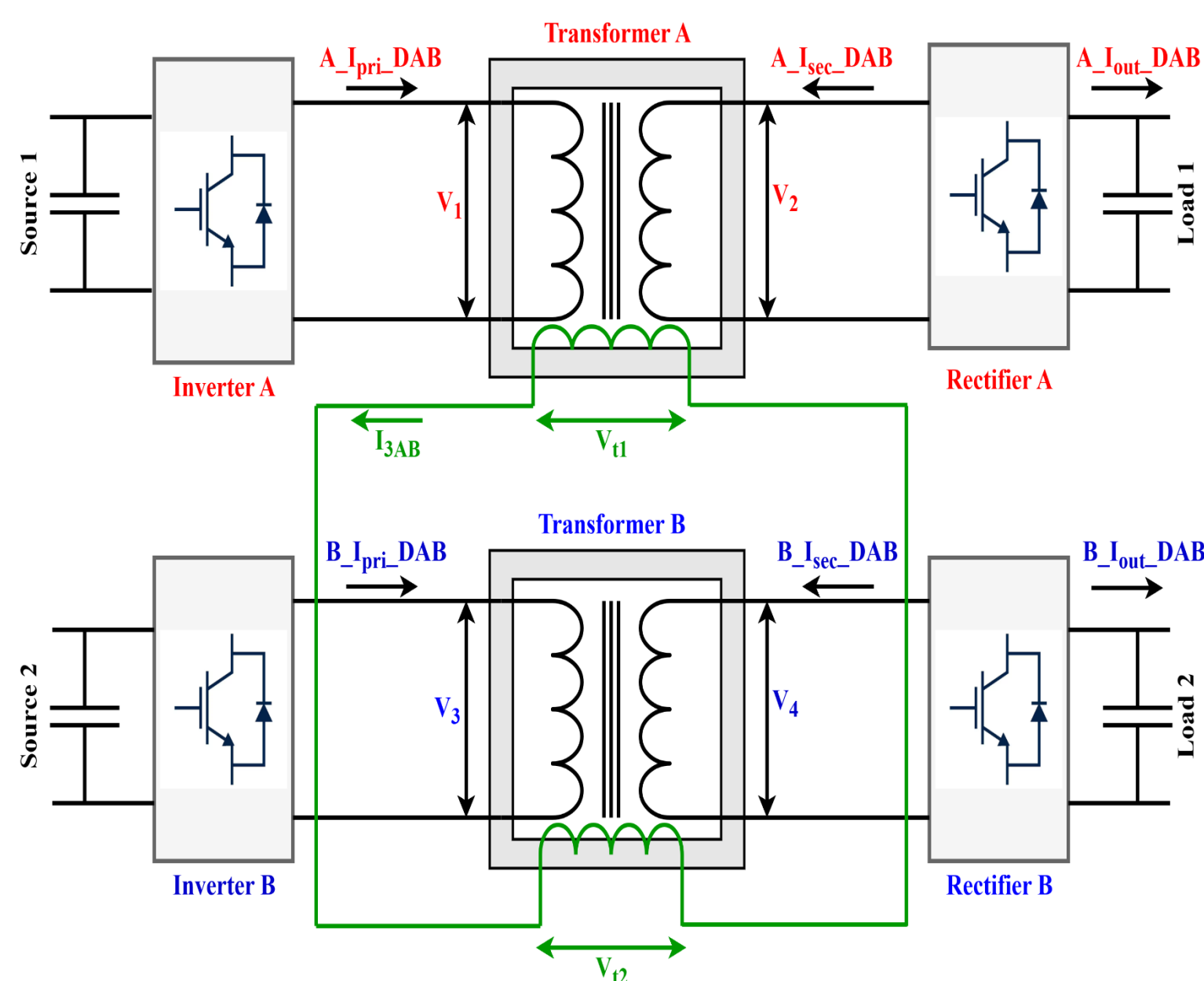


Figure 2: Quad-Port Dual Active Bridge Converter

- ❖ The control begins with single phase shift modulation, where there will be a relative phase shift between the two active bridges. It is simple but encourage reactive current. Dual phase shift modulation adds a second, inner phase shift inside each bridge, creating brief zero-level intervals in the bridge voltage. Those intervals let the controller represent the inductor current, shorten the non-productive circulation, and widen the range over which soft switching is maintained.

3. RESULTS AND DISCUSSIONS

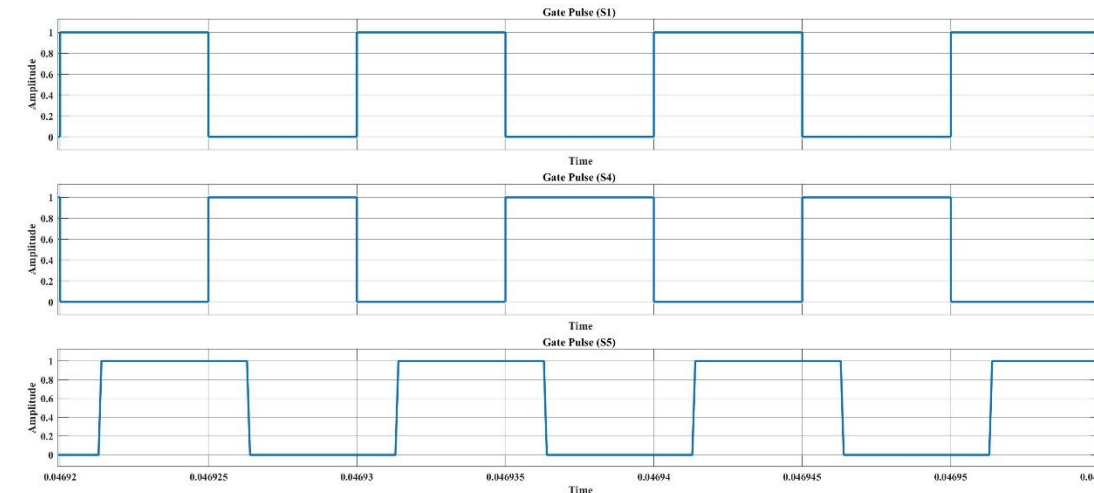


Figure 3 (a): Gate Pulses S1,S4,S5 for SPS DAB

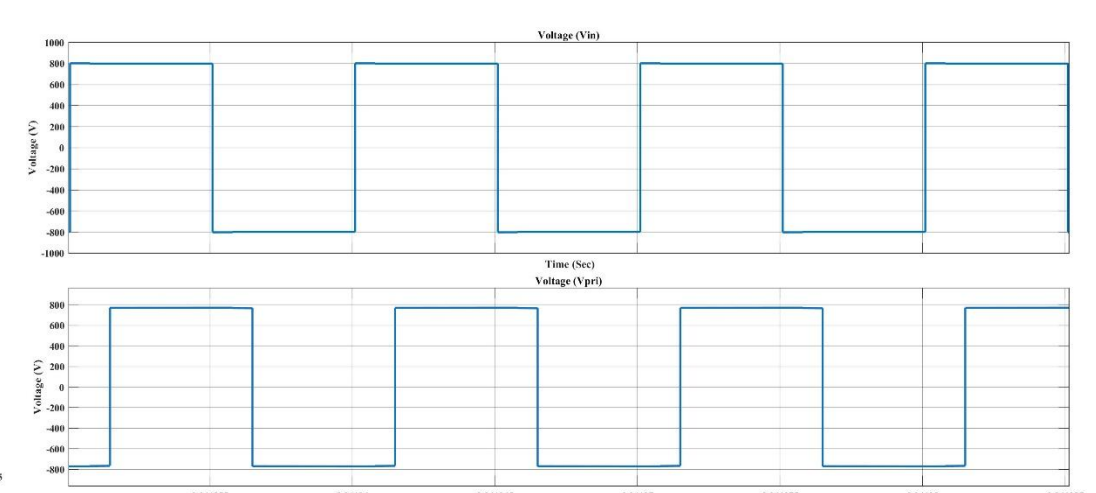


Figure 3 (b): Voltage Vin,Vpri for SPS DAB

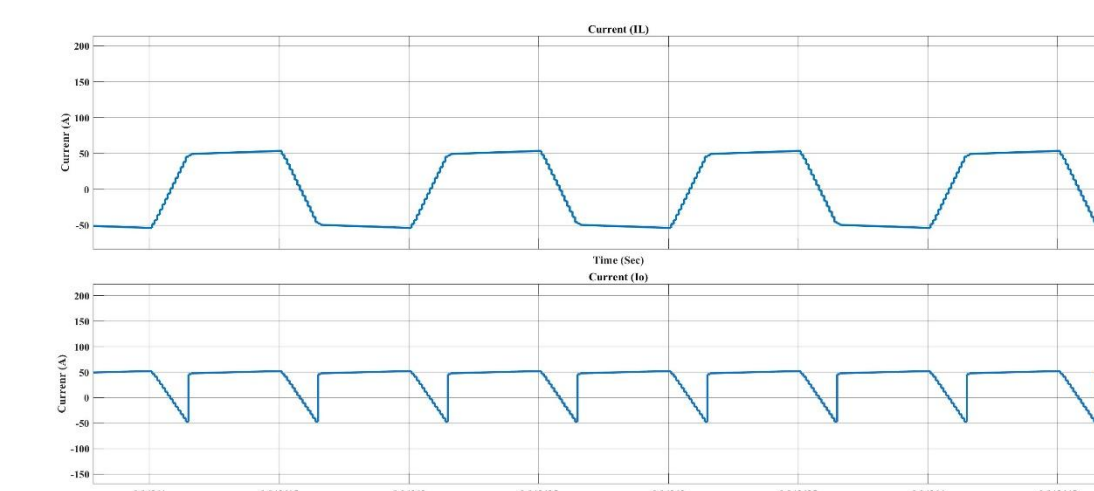


Figure 3 (c): Current IL,Io for SPS DAB

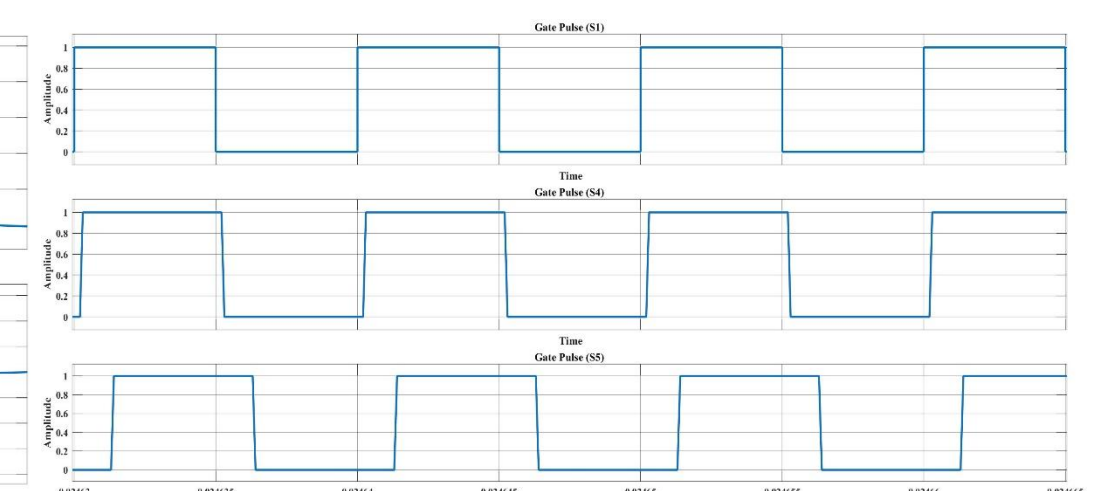


Figure 4 (a): Gate Pulses S1,S4,S5 for DPS DAB

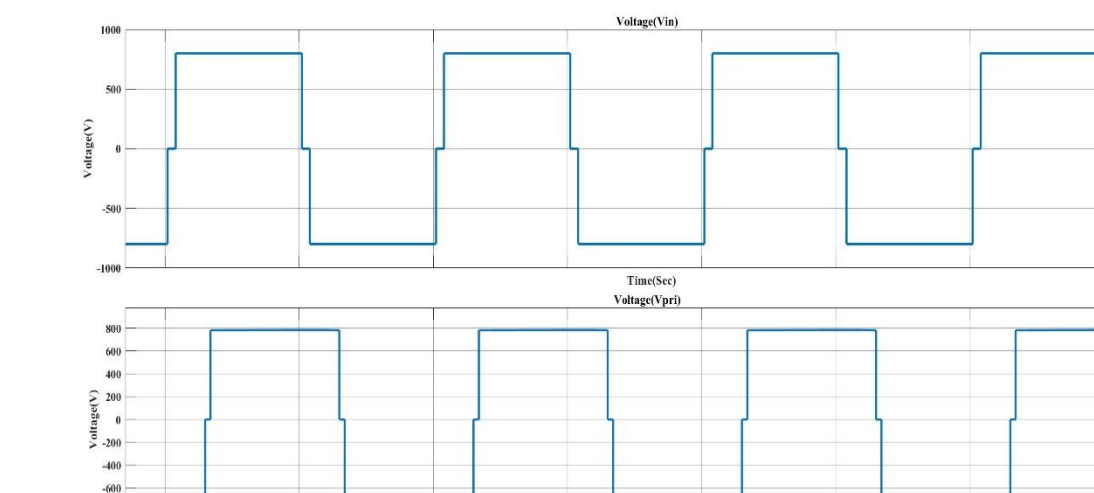


Figure 4 (b): Voltage Vin,Vpri for DPS DAB

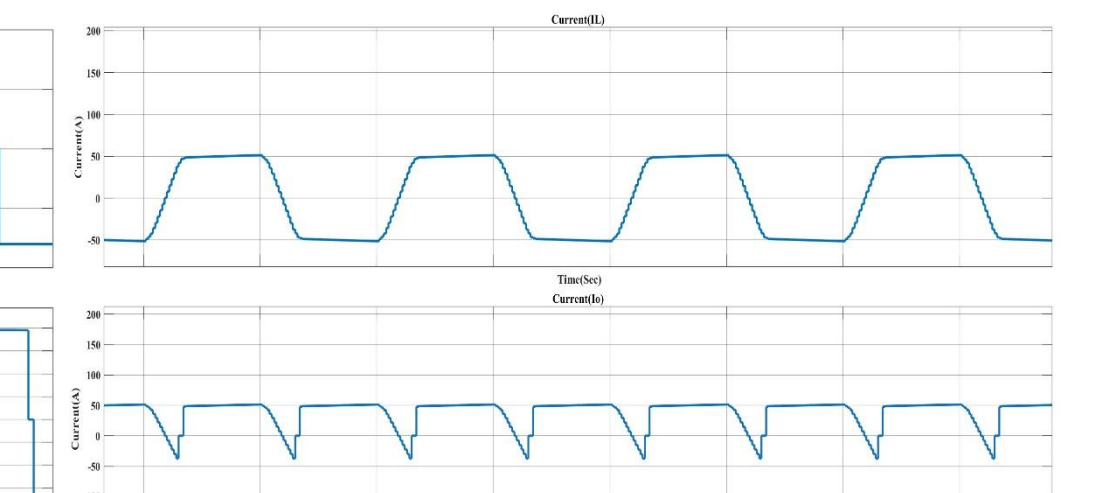


Figure 4 (c): Current IL,Io for DPS DAB

The DAB power stage uses two full H-bridges and a high-frequency transformer, the transformer's leakage inductance $L_s=21.33 \mu\text{H}$ is the energy transfer path. The simulation is set at $V_1=V_2=800\text{V}$, $f_s=100 \text{ kHz}$, turns ratio $n=1$, with a 30KW operating point.

With Single Phase Shift (SPS), only the relative phase is commanded and phase shift between bridges of gate pulse can be seen, the power flow happens. Then compared with dual phase shift (DPS) of dual active bridge converter where two phase shifts were provided in which the inner phase shift and outer phases shift of gate pulses can be seen, also the nature of inductor current and output current can be seen, this modulation technique is followed to reduce the RMS value of the current flowing through the inductor.

4. FUTURE WORK

- ❖ Future work will focus on Goal-2 ie., power stage analysis of a three winding, quad port dual active bridge converter. We will apply the same methodology used for the two winding DAB, then implement the three winding quad port dual active bridge in hardware to validate performance under real operating conditions. A closed loop controller will be developed to improve voltage regulation and dynamic response. Finally, adding bidirectional charging capability will make the system suitable for future smart grid environments.

5. REFERENCES

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